BQ78350-R1A

Technical Reference

TEXAS INSTRUMENTS

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Contents

Preface

Read this First

This manual discusses the modules and peripherals of the BQ78350-R1A device, and how each is used to build a complete battery pack gas gauge and protection solution. For electrical specifications, see the *BQ78350-R1A CEDV Li-Ion Gas Gauge and Battery Management Controller Companion to the BQ769x0 Battery Monitoring AFE Data Sheet* (SLUSE05).

Notational Conventions

The following notational conventions are used if SBS commands and data flash values are mentioned within a text block:

- SBS commands: *italics* with parentheses and no breaking spaces; for example, *RemainingCapacity()*
- Data flash: *italics*, **bold**, and breaking spaces; for example, ***Design Capacity***
- Register bits and flags: *italics* and brackets; for example, *[TDA]*
- Data flash bits: *italics* and **bold**; for example, ***[LED1]***
- Modes and states: ALL CAPITALS; for example, UNSEALED

The reference format for SBS commands is: SBS:Command Name(Command No.): Manufacturer Access(MA No.)[Flag], for example:

SBS:Voltage(0x09), or SBS:ManufacturerAccess(0x00): Seal Device(0x0020)

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Introduction

The BQ78350-R1A device provides a feature-rich battery management solution for 3-series cell to 15-series cell battery pack applications. The device has extended capabilities, including:

- Companion Protection Controller to the BQ76920, BQ76930, and BQ76940 AFE Devices for Li-Ion or LiFePO4 Battery Packs
- Compensated End-of-Discharge Voltage (CEDV) Gas Gauging Algorithm Accurately Measures Available Charge and State-of-Health
- Voltage Based Cell Balancing Control
- Normal and Lower Power Modes
  - NORMAL
  - SLEEP
  - SHUTDOWN
- Full Array of Programmable Protection Features
  - Voltage
  - Current
  - Temperature
  - Charge Timeout
  - CHG/DSG FETs
- Precharge and Fast Charge Algorithm
- Diagnostic Lifetime Data Monitor
- Black Box Event Recorder
- Supports Two-Wire SMBus v1.1 Interface
- SHA-1 Authentication
- Package: 30-Lead TSSOP

The BQ78350-R1A is intended to be used with the BQ769x0 Battery Monitor with a 2.5-V REGOUT configuration and I^{2}C Address 0x08. However, the BQ78350-R1A can use a BQ769x0 with or without the communications CRC enabled (the BQ78350-R1A automatically detects if CRC is enabled).

Basic Measurement System

2.1 Introduction

NOTE: For this section, refer to the *BQ769x0 3-Series to 15-Series Cell Battery Monitor Family for Li-Ion and Phosphate Applications Data Manual* (SLUSBK2) for further details.

The BQ78350-R1A reads the BQ769x0 companion AFE registers that contain recent values from the integrating analog-to-digital converter (ADC) for current measurement, and a second delta-sigma ADC for individual cell and temperature measurements. The BQ78350-R1A also has the capability to measure the battery voltage through an externally translated voltage.

2.2 Current and Coulomb Counting

The integrating delta-sigma ADC in the companion BQ769x0 AFE measures the charge/discharge flow of the battery by measuring the voltage drop across a small-value sense resistor between the SRP and SRN pins. The 16-bit integrating ADC measures bipolar signals from –0.20 V to 0.20 V with 8.44-µV resolution. The AFE reports charge activity when VSR = $V_{(SRP)} - V_{(SRN)}$ is positive, and discharge activity when VSR = $V_{(SRP)} - V_{(SRN)}$ is negative. The BQ78350-R1A continuously monitors the measured current and integrates the digital signal from the AFE over time using an internal counter.

To support large battery configurations, the current data can be scaled to ensure accurate reporting through the SMBus. The data reported is scaled based on the setting of the *SpecificationInfo()* command.

The data reported through the *Current()* can also have a deadband applied to it. This removes any noise or offset that has not been calibrated out from being reported as real current. This value is programmed in ***Deadband*** with a default configured for mA scaling in *SpecificationInfo()*. If the *SpecificationInfo()* IPSCALE is set to 10x or 100x, then it is strongly recommended to set ***Deadband*** to 1.

2.3 Voltage

The BQ78350-R1A updates the individual series cell voltages through the BQ769x0 at 250-ms intervals. The BQ78350-R1A configures the BQ769x0 to connect to the selected cells in sequence and uses this information for cell balancing and individual cell fault functions. The internal 14-bit ADC of the BQ769x0 measures each cell voltage value, which is then communicated digitally to the BQ78350-R1A where it is scaled and translated into unit millivolts. The maximum supported input range of the ADC is 6.075 V.

The BQ78350-R1A also separately measures the average cell voltage through an external translation circuit at the BAT pin. This value is specifically used for the gas gauge algorithm. The external translation circuit is controlled via the VEN pin so that the translation circuit is only enabled when required to reduce overall power consumption. VEN requires an external pullup to VCC, typically 100 k, to operate correctly.

In addition to the voltage measurements used by the BQ78350-R1A algorithms, there is an optional auxiliary voltage measurement capability via the VAUX pin. This feature measures the input on a 250-ms update rate and provides the programmable scaled value through the *VAUXVoltage()* SMBus command. The data can be enabled to influence selected fault recovery features. See *General Protections Configuration*, ***[VAUXR]***, for further details.

The VEN pin will go high 2 ms prior to the BAT being measured if ***DA Configuration [ExtAveEN]*** = 1, and then return low unless ***DA Configuration [VAUXEN]*** = 1, which will cause VEN to remain high for a further 2 ms prior to making the VAUX measurement. This results in VEN possibly being high for up to 40 ms per second in NORMAL mode.

To support large battery configurations where the battery voltage can exceed 32767 mV, the data should be scaled as the gauge's internal data processing is done in a signed integer range (–32768 to 32767) to ensure accurate reporting through the SMBus. The data reported is scaled based on the setting of the *SpecificationInfo()* command. The cell voltages are not scaled.

2.4 Temperature

The BQ78350-R1A receives temperature information from external or internal temperature sensors in the BQ769x0 AFE. Depending on the number of series cells supported, the AFE will provide one, two, or three external thermistor measurements. The value of temperature is reported through *Temperature()* and can be configured in DA Configuration.

2.4.1 FET Temperature Measurement

The BQ78350-R1A can be configured to report FET temperature, which can be available through *DAStatus2()*. If multiple temperature sensors are selected for FET temperature, then either the average or highest is used based on the setting of ***[FTEMP]*** in ***[DA Configuration]***.

The selection of temperature sensor as cell temperature protection or FET temperature protection can be made through the ***Temperature Mode*** register.

2.4.2 Temperature Enable

This register enables/disables the available temperature sensor options.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	Temperature Enable	hex	1	0x00	0xFF	0x09	—

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Low Byte	RSVD	RSVD	RSVD	RSVD	SOURCE	TS3	TS2	TS1

RSVD (Bits 7–4): Reserved

SOURCE (Bit 3): Configure the use of Internal or External temperature sensors for all AFE ports

0 = Use internal temperature sensor(s)

1 = Use external temperature sensor(s)

TS3 (Bit 2): Enable/disable companion AFE temperature sensor TS3, if available

0 = Disable TS3 temperature sensor

1 = Enable TS3 temperature sensor

TS2 (Bit 1): Enable/disable companion AFE temperature sensor TS2, if available

0 = Disable TS2 temperature sensor

1 = Enable TS2 temperature sensor

TS1 (Bit 0): Enable/disable companion AFE temperature sensor TS1

0 = Disable TS1 temperature sensor

1 = Enable TS1 temperature sensor

2.4.3 Temperature Mode Configuration

Each available external temperature sensor can be configured to be used for the cell temperature or FET temperature features.

Figure 2-1. Temperature Mode

7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD	RSVD	RSVD	TSMode3	TSMode2	TSMode1

RSVD (Bits 7–3): Reserved

TSMode3 (Bit 2): Select TS3 sensor for Cell or FET Temperature Protection
- 0 = Use for Cell (default)
- 1 = Use for *FETTemperature()*

TSMode2 (Bit 1): Select TS2 sensor for Cell or FET Temperature Protection
- 0 = Use for Cell (default)
- 1 = Use for *FETTemperature()*

TSMode1 (Bit 0): Select TS1 sensor for Cell or FET Temperature Protection
- 0 = Use for Cell (default)
- 1 = Use for *FETTemperature()*

2.5 Temperature Ranges

The measured temperature is segmented into several temperature ranges. The BQ78350-R1A uses these as indication, and, for Lifetime Data Logging, the time spent in each range. The temperature ranges set in data flash should adhere to the following format:

T1 ≤ T2 ≤ T3 ≤ T4

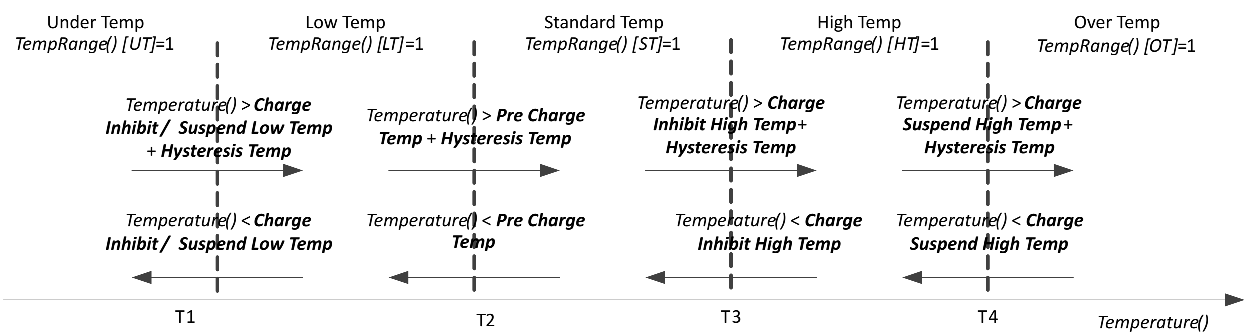


Figure 2-2. Data Flash Temperature Range Format

See the Temperature Ranges data flash subclass for details on the specific data flash variables.

2.6 Basic Configuration Options

There are a variety of options available in the BQ78350-R1A and the companion AFE that influence the startup conditions, system configuration, and the data measurement system.

2.6.1 DA Configuration

This register is used to configure the setup of various measurement features of the BQ78350-R1A.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	DA Configuration	hex	1	0x00	0xFF	0x11	—

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Low Byte	FTEMP	CTEMP	RSVD	ExtAveEN	VAUXEN	VAUX_SCALE	IN_SYSTEM_SLEEP	SLEEP

FTEMP (Bit 7): FET Temperature Protection Source

0 = Maximum of external available sources (default)

1 = Average of external available sources

CTEMP (Bit 6): Cell Temperature Protection Source

0 = Maximum of external available sources (default)

1 = Average of external available sources

RSVD (Bit 5): Reserved

ExtAveEN (Bit 4): Enables the BQ78350-R1A to measure the BAT input

0 = BAT input is not measured.

1 = BAT input is measured and made available via *ExtAveCellVoltage()* (default).

VAUXEN (Bit 3): Enables the BQ78350-R1A to measure the VAUX input

0 = VAUX input is not measured (default).

1 = VAUX input is measured and made available via *VAUXVoltage().*

VAUX_SCALE (Bit 2): Enables the BQ78350-R1A to scale the *VAUXVoltage()* data by 10. For example: Units are 10 mV rather than 1 mV.

0 = *VAUXVoltage()* is not scaled (resolution is 1 mV) (default).

1 = *VAUXVoltage()* is scaled (resolution is 10 mV).

IN_SYSTEM_SLEEP (Bit 1): IN SYSTEM SLEEP mode

1 = Enable

0 = Disable (default)

SLEEP (Bit 0): Enables the BQ78350-R1A to enter SLEEP mode.

0 = The BQ78350-R1A never enters SLEEP mode.

1 = The BQ78350-R1A enters SLEEP mode under normal sleep entry criteria (default).

2.6.2 FET Options

This register configures the various FET control options.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	FET Options	hex	2	0x0000	0xFFFF	0x0021	—

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
High Byte	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	KEY_POL	PCHG_POL
Low Byte	RSVD	SLEEPCHG	CHGFET	CHGIN	CHGSU	OTFET	KEY_EN	PCHG_EN

RSVD (Bits 7–2): Reserved

KEY_POL: This bit configures the KEYIN input detection polarity.

0 = KEYIN detection is active low (default).

1 = KEYIN detection is active high.

PCHG_POL: Configures the BQ78350-R1A $\overline{\text{PRECHG}}$ pin output polarity. If PCHG_EN = 0, then this bit has no influence.

0 = The BQ78350-R1A configures the $\overline{\text{PRECHG}}$ as active low (default).
1 = The BQ78350-R1A configures the $\overline{\text{PRECHG}}$ as active high, requiring an external pullup.

SLEEPCHG: CHG FET is enabled during SLEEP.
0 = CHG FET off during SLEEP (default).
1 = CHG FET remains on during SLEEP.

CHGFET: FET action on valid charge termination
0 = FET active
1 = Charging and Precharging disabled, FET off (default)

CHGIN: FET action in CHARGE INHIBIT mode
0 = FET active (default)
1 = Charging and Precharging disabled, FETs off

CHGSU: FET action in CHARGE SUSPEND mode
0 = FET active (default)
1 = Charging and Precharging disabled, FETs off

OTFET: FET action in OVERTEMPERATURE mode
0 = No FET action for overtemperature condition (default)
1 = CHG and DSG FETs will be turned off for overtemperature conditions.

KEY_EN: Enables the BQ78350-R1A to use the KEYIN pin function
0 = The BQ78350-R1A never uses KEYIN (default).
1 = The BQ78350-R1A KEYIN is used to control the DSG FET.

PCHG_EN: Enables the BQ78350-R1A to use the $\overline{\text{PRECHG}}$ pin during PRECHARGE mode
0 = The BQ78350-R1A never uses $\overline{\text{PRECHG}}$.
1 = The BQ78350-R1A controls $\overline{\text{PRECHG}}$ under normal charge control algorithm (default).

2.6.3 AFE Cell Map

This register maps the cells connected to the companion AFE so that the BQ78350-R1A knows cells are present at the indicated VCx channel.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Configuration	AFE	AFE Cell Map	hex	2	0x0000	0xFFFF	0x0013	—

	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
High Byte	RSVD	VC15	VC14	VC13	VC12	VC11	VC10	VC9
Low Byte	VC8	VC7	VC6	VC5	VC4	VC3	VC2	VC1

RSVD (Bit 7): Reserved

VCx: Cell connected to this node
1 = A cell is connected to this node and valid measurements are expected.
0 = A cell is NOT connected to this node.

The BQ78350-R1A determines which companion AFE is connected by the total number of cells connected.

- When Series Cells = 3 to 5, the BQ76920 companion AFE is used.
- When Series Cells = 6 to 10, the BQ76930 companion AFE is used.
- When Series Cells = 9 to 15, the BQ76940 companion AFE is used.

Protections

3.1 Introduction

The BQ78350-R1A supports a wide range of battery and system protection features that are easily configured or enabled via the integrated data flash. All of the protection items can be enabled or disabled under ***Settings:Enable Protections A***, ***Settings:Enable Protections B***, and ***Settings:Enable Protections C***.

If the CHG FET is off and the gauge detects discharge current ≥ ***Dsg Current Threshold***, then the CHG FET is turned on to protect CHG FET body diode. The CHG FET is turned back off once discharge current is removed. If the DSG FET is off and the gauge detects charge current ≥ ***Chg Current Threshold***, then the DSG FET is turned on to protect the DSG FET body diode. The DSG FET is turned back off once charge current is removed. Body diode protection is always active.

3.1.1 General Protections Configuration

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Protection	Protection Configuration	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
RSVD	RSVD	CC_DSG_OFF	DC_CHG_OFF	LPEN	VAUXR	CUV_RECOV_CHG	RSVD

RSVD (Bits 7–6): Reserved

CC_DSG_OFF (Bit 5): Turns DSG FET OFF in current-based charge faults

0 = Disabled (default)
1 = Enabled

DC_CHG_OFF (Bit 4): Turns CHG FET OFF in current-based discharge faults

0 = Disabled (default)
1 = Enabled

LPEN (Bit 3): Protection recovery uses the LOAD_PRESENT flag in the AFE to determine discharge fault recovery. LOAD_PRESENT should only be used in a low-side protection FET configuration.

0 = Disabled (default)
1 = Enabled

VAUXR (Bit 2): Protection recovery uses the VAUX input as charger present detection.

0 = Disabled (default)
1 = Enabled

CUV_RECOV_CHG (Bit 1): Requires charge to recover *SafetyStatus()[CUV]*

0 = Disabled (default)
1 = Enabled

RSVD (Bit 0): Reserved

3.1.2 Enabled Protections

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Protection	Enabled Protections A	Hex	1	0x00	0xFF	0xFF	—

7	6	5	4	3	2	1	0
ASCDL	ASCD	AOLDL	AOLD	OCD	OCC	COV	CUV

ASCDL (Bit 7): Short Circuit in Discharge Latch
- 0 = Disabled
- 1 = Enabled

ASCD (Bit 6): Short Circuit in Discharge recovery. Detection of an ASCD fault cannot be disabled.
- 0 = Bypassed, auto recovers within 250 ms
- 1 = Enabled

AOLDL (Bit 5): Overload in Discharge Latch
- 0 = Disabled
- 1 = Enabled

AOLD (Bit 4): Overload in Discharge recovery. Detection of an AOLD fault cannot be disabled.
- 0 = Bypassed, auto recovers within 250 ms
- 1 = Enabled

OCD (Bit 3): Overcurrent in Discharge
- 0 = Disabled
- 1 = Enabled

OCC (Bit 2): Overcurrent in Charge
- 0 = Disabled
- 1 = Enabled

COV (Bit 1): Cell Overvoltage
- 0 = Disabled
- 1 = Enabled

CUV (Bit 0): Cell Undervoltage
- 0 = Disabled
- 1 = Enabled

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Protection	Enabled Protections B	Hex	1	0x00	0xFF	0x0F	—

7	6	5	4	3	2	1	0
RSVD	OCDL	OTF	AFE_OVRD	UTD	UTC	OTD	OTC

RSVD (Bit 7): Reserved

OCDL (Bit 6): Overcurrent in Discharge Latch
- 0 = Disabled (default)
- 1 = Enabled

OTF (Bit 5): Overtemperature Fault
- 0 = Disabled (default)
- 1 = Enabled

AFE_OVRD (Bit 4): AFE ALERT
- 0 = Disabled (default)
- 1 = Enabled

UTD (Bit 3): Undertemperature in Discharge
- 0 = Disabled
- 1 = Enabled

UTC (Bit 2): Undertemperature in Charge
- 0 = Disabled
- 1 = Enabled

OTD (Bit 1): Overtemperature in Discharge
- 0 = Disabled
- 1 = Enabled

OTC (Bit 0): Overtemperature in Charge
- 0 = Disabled
- 1 = Enabled

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Protection	Enabled Protections C	Hex	1	0x00	0xFF	0x15	—

7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD	OC	CTOS	CTO	PTOS	PTO

RSVD (Bits 7–5): Reserved. Do not use.

OC (Bit 4): Overcharge
- 0 = Disabled
- 1 = Enabled

CTOS (Bit 3): Charging Timeout Suspended
- 0 = Disabled
- 1 = Enabled

CTO (Bit 2): Charging Timeout
- 0 = Disabled

1 = Enabled

PTOS (Bit 1): Precharging Timeout Suspend

0 = Disabled

1 = Enabled

PTO (Bit 0): Precharging Timeout

0 = Disabled

1 = Enabled

3.1.3 Enabled Removal Recovery

The BQ78350-R1A offers the option to recover current-based protection by detecting the $\overline{\text{PRES}}$ pin transition from high to low; for example, the pack is removed and re-inserted into the system.

To enable the replacement recovery, the appropriate bit in ***Enable Removable Recovery A and Enable Removable Recovery B*** should be set. When the bit is set, then the high to low transition of $\overline{\text{PRES}}$ becomes the only recovery method.

Table 3-1. Enabled Removal Recovery A

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Protection	Enable Removable Recovery A	Hex	1	0x00	0xff	0x00	—

7	6	5	4	3	2	1	0
ASCDL	ASCD	AOLDL	AOLD	OCD	OCC	RSVD	RSVD

ASCDL (Bit 7): ASCDL Protection Removal recovery

0 = Standard recovery only enabled (default)

1 = Removal recovery only enabled

ASCD (Bit 6): ASCD Protection Removal recovery

0 = Standard recovery only enabled (default)

1 = Removal recovery only enabled

AOLDL (Bit 5): AOLDL Protection Removal recovery

0 = Standard recovery only enabled (default)

1 = Removal recovery only enabled

AOLD (Bit 4): AOLD Protection Removal recovery

0 = Standard recovery only enabled (default)

1 = Removal recovery only enabled

OCD (Bit 3): OCD Protection Removal recovery

0 = Standard recovery only enabled (default)

1 = Removal recovery only enabled

OCC (Bit 2): OCC Protection Removal recovery

0 = Standard recovery only enabled (default)

1 = Removal recovery only enabled

RSVD (Bits 1–0): Reserved. Do not use.

Table 3-2. Enabled Removal Recovery B

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Protection	Enable Removable Recovery B	Hex	1	0x00	0xff	0x00	—

7	6	5	4	3	2	1	0
RSVD	OCDL	RSVD	RSVD	RSVD	RSVD	RSVD	RSVD

RSVD (Bit 7): Reserved. Do not use.

OCDL (Bit 6): OCDL Protection Removal recovery

0 = Standard recovery only enabled (default)

1 = Removal recovery only enabled

RSVD (Bits 5–0): Reserved. Do not use.

3.1.4 FET Action Options for Current Protections

The BQ78350-R1A offers the option to turn off the CHG FET during an overcurrent in discharge (OCD), overcurrent in discharge latch (OCDL), overload (AOLD), overload latch (AOLDL) or short circuit in discharge (ASCD), short circuit in discharge latch (ASCDL) faults, or the DSG FET in overcurrent in charge (OCC) faults.

The CHG FET will turn off for the OCD, OCDL, AOLD, AOLDL, ASCD, and ASCDL faults when ***[DC_CHG_OFF]*** in ***Protection Configuration*** is set.

The DSG FET will turn off for the OCC faults when ***[CC_DSG_OFF]*** in ***Protection Configuration*** is set.

3.2 Cell Undervoltage Protection

The device can detect undervoltage in batteries and protect cells from damage by preventing further discharge.

Upon CUV detection, a snapshot of the measured cell voltages are made available in *CUVSnapshot()*. This snapshot is available until the next instance of a CUV fault, as this causes the data to be updated to the latest set of measurements.

Status	Condition	Action
Normal	All Cell voltages in *CellVoltage1..15()* > ***CUV:Threshold***	*SafetyAlert()[CUV]* = 0 *BatteryStatus()[TDA]* = 0
Alert	Any Cell voltages in *CellVoltage1..15()* ≤ ***CUV:Threshold***	*SafetyAlert()[CUV]* = 1 *BatteryStatus()[TDA]* = 1
Trip	Any Cell voltages in *CellVoltage1..15()* ≤ ***CUV:Threshold*** for ***CUV:Delay*** duration	*SafetyAlert()[CUV]* = 0 *SafetyStatus()[CUV]* = 1 *BatteryStatus()[FD]* = 1 *OperationStatus()[XDSG]* = 1 Discharging is not allowed.
Recovery	*SafetyStatus()[CUV]* = 1 AND All Cell voltages in *CellVoltage1..15()* ≥ ***CUV:Recovery*** AND ***Protection Configuration[CUV_RECOV_CHG]*** = 0 OR ***[CUV_RECOV_CHG]*** = 1 AND Charging detected (that is, *BatteryStatus[DSG]* = 0)	*SafetyStatus()[CUV]* = 0 *BatteryStatus()[FD]* = 0, *[TDA]* = 0 *OperationStatus()[XDSG]* = 0 Discharging is allowed.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	CUV	Threshold	I2	0	5000	2500	mV
Protections	CUV	Delay	U1	0	255	2	s
Protections	CUV	Recovery	I2	0	5000	3000	mV

3.3 Cell Overvoltage Protection

The device can detect cell overvoltage in batteries and protect cells from damage by preventing further charging.

Upon COV detection, a snapshot of the measured cell voltages are made available in *COVSnapshot()*. This snapshot is available until the next instance of a COV fault, as this causes the data to be updated to the latest set of measurements.

Status	Condition	Action
Normal	All voltages in *CellVoltage1..15()* < ***COV:Threshold***	*SafetyAlert()[COV]* = 0
Alert	Any voltage in *CellVoltage1..15()* ≥ ***COV:Threshold***	*SafetyAlert()[COV]* = 1 *BatteryStatus()[TCA]* = 1
Trip	Any voltage in *CellVoltage1..15()* ≥ ***COV:Threshold*** continuous ≥ ***COV:Threshold*** for ***COV:Delay*** duration	*SafetyAlert()[COV]* = 0 *SafetyStatus()[COV]* = 1 *BatteryStatus()[TCA]* = 0
Recovery	*SafetyStatus()[COV]* = 1 AND ***Protection Configuration:VAUXR*** = 0 all voltages in *CellVoltage1..15()* ≤ ***COV:Recovery***	*SafetyStatus()[COV]* = 0 *BatteryStatus()[TCA]* = 0
Recovery	*SafetyStatus()[COV]* = 1 AND ***Protection Configuration:VAUXR*** = 1 all voltages in *CellVoltage1..15()* ≤ ***COV:Recovery*** AND *VAUXVoltage()* < ***Power:Charger Present Threshold***	*SafetyStatus()[COV]* = 0 *BatteryStatus()[TCA]* = 0

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	COV	Threshold	I2	0	32767	4300	mV
Protections	COV	Delay	U1	0	255	2	s
Protections	COV	Recovery	I2	0	32767	4100	mV

3.4 Overcurrent in Charge Protection

The device has overcurrent in charge protection that can be configured to specific current and delay thresholds to accommodate charging behaviors. See Section 3.1.4 for additional FET action options.

Status	Condition	Action
Normal	*Current()* < ***OCC:Threshold***	*SafetyAlert()[OCC]* = 0
Alert	*Current()* ≥ ***OCC:Threshold***	*SafetyAlert()[OCC]* = 1 *BatteryStatus()[TCA]* = 1
Trip	*Current()* continuous ≥ ***OCC:Threshold*** for ***OCC:Delay*** duration	*SafetyAlert()[OCC]* = 0 *SafetyStatus()[OCC]* = 1 *BatteryStatus()[TCA]* = 0 *OperationStatus()[XCHG]* = 1 Charging is not allowed.
Recovery	*[SafetyStatus()[OCC]* = 1 AND *Current()* continuous ≤ ***OCC:Recovery Threshold*** for ***OCC:Recovery Delay*** time	*SafetyStatus()[OCC]* = 0 *BatteryStatus()[TCA]* = 0 *OperationStatus()[XCHG]* = 0 Charging is allowed.

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	OCC	Threshold	I2	–32768	32767	6000	mA	Overcurrent in Charge trip threshold
Protections	OCC	Delay	U1	0	255	6	s	Overcurrent in Charge trip delay
Protections	OCC	Recovery Threshold	I2	–32768	32767	–200	mA	Overcurrent in Charge recovery threshold
Protections	OCC	Recovery Delay	U1	0	255	5	s	Overcurrent in Charge recovery delay

3.5 Overcurrent in Discharge Protection

The device has two independent overcurrent in discharge protections that can be set to different current and delay thresholds to accommodate different load behaviors. See Section 3.1.4 for additional FET action options.

Status	Condition	Action
Normal	*Current()* > ***OCD:Threshold***	*SafetyAlert()[OCDL]* = 0, if OCDL counter = 0
Alert	OCDL counter > 0	*SafetyAlert()[OCDL]* = 1, Decrement OCDL counter by one after each ***OCD:Counter Dec Delay*** period
Trip	*Current()* continuous ≤ ***OCD:Threshold*** for ***OCD:Delay*** duration	*SafetyAlert()[OCD]* = 0 *SafetyStatus()[OCD]* = 1 *OperationStatus()[XDSG]* = 1 DSG FET is disabled. Increment OCDL counter.
Latch	OCDL counter ≥ ***OCD:Latch Limit***	*SafetyAlert()[OCDL]* = 0 *SafetyStatus()[OCDL]* = 1 *OperationStatus()[XDSG]* = 1 DSG FET is disabled.
Recovery	*[SafetyStatus()[OCD]* = 1 AND ***Protection Configuration:VAUXR*** = 0 *Current()* continuous ≥ ***OCD:Recovery Threshold for OCD:Recovery Delay*** time	*SafetyStatus()[OCD]* = 0 *OperationStatus()[XDSG]* = 0 Discharging is allowed.
Recovery	*[SafetyStatus()[OCD]* = 1 AND ***Protection Configuration:VAUXR*** = 1 *Current()* continuous ≥ ***OCD:Recovery Threshold*** for ***OCD:Recovery Delay*** time OR *VAUXVoltage()* ≥ ***Power:Charger Present Threshold***	*SafetyStatus()[OCD]* = 0 *OperationStatus()[XDSG]* = 0 Discharging is allowed.
Latch Reset	*SafetyStatus()[OCDL]* = 1 for OCD: Reset Time	*SafetyStatus()[OCDL]* = 0 Reset OCDL counter *OperationStatus()[XDSG]* = 0 DSG FET returns to normal if *SafetyStatus()[OCD]* = 0.

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
Protections	OCD	Threshold	I2	–32768	32767	–6000	mA	Overcurrent in Discharge trip threshold
Protections	OCD	Delay	U1	0	255	6	s	Overcurrent in Discharge trip delay
Protections	OCD	Recovery Threshold	I2	–32768	32767	200	mA	Overcurrent in Discharge recovery threshold
Protections	OCD	Recovery Delay	U1	0	255	5	s	Overcurrent in Discharge recovery delay
Protections	OCDL	Latch Limit	U1	0	255	0	counts	Overcurrent in Discharge latch limit
Protections	OCDL	Counter Dec Delay	U1	0	255	10	s	Overcurrent in Discharge counter decrement delay
Protections	OCDL	Reset	U1	0	255	15	s	Overcurrent in Discharge latch reset delay

3.6 Hardware-Based Protection

The BQ78350-R1A device has two main hardware-based protections, AOLD and ASCD, with adjustable current and delay time. Setting ***ASCD Threshold and Delay [RSNS]*** doubles the threshold value. It is located in bit 8 of the ***ASCD Threshold Delay*** register. The ***Threshold*** settings are in mV; therefore, the actual current that triggers the protection is based on the R_{SENSE} used in the schematic design.

For details on how to configure the AFE hardware protection, refer to the tables in the companion data manual, *BQ769x0 3-Series to 15-Series Cell Battery Monitor Family for Li-Ion and Phosphate Applications* (SLUSBK2).

All of the hardware-based protections provide a short term Trip/Alert/Recovery protection to account for a current spike as well as a Trip/Alert/Latch protection for persistent faulty condition. The latch feature also stops the FETs from toggling on and off continuously, preventing damage to the FETs.

In general, when a fault is detected after the ***Delay*** time, the DSG FET will be disabled. However, if ***Protection Configuration [LPEN]*** is set, then both FETs are turned off (Trip stage), and an internal fault counter will be incremented (Alert stage). As the DSG FET is turned off, the current will drop to 0 mA. After ***Recovery*** time, the CHG and DSG FETs will be turned on again (Recovery stage) unless additional recovery conditions are enabled.

If the alert is caused by a current spike, the fault count will be decremented after ***Counter Dec Delay*** time. If this is a persistent faulty condition, the device will enter the Trip stage after ***Delay*** time, and repeat the Trip/Alert/Recovery cycle. The internal fault counter is incremented every time the device goes through the Trip/Alert/Recovery cycle. Once the internal fault counter hits the ***Latch Limit***, the protection enters a Latch stage and the fault will only be cleared through the Latch Reset condition. If Latch Limit is set to 0, it will latch after the first detection.

The Trip/Alert/Recovery/Latch stages are documented in each of the following hardware-based protection sections.

3.6.1 Overload in Discharge Protection

The device has a hardware-based overload in discharge protection with adjustable current and delay. See Section 3.1.4 for additional FET action options.

Status	Condition	Action
Normal	*Current()* > (***AOLD Threshold and Delay[3:0]***/R_{SENSE})	*SafetyAlert()[AOLDL]* = 0, if AOLDL counter = 0
Alert	AOLDL counter > 0	*SafetyAlert()[AOLDL]* = 1 Decrement AOLDL counter by one after each ***AOLD:Counter Dec Delay*** period
Trip	*Current()* continuous ≤ (***AOLD Threshold and Delay[3:0]***/R_{SENSE}) for ***AOLD Threshold and Delay[6:4]*** duration	*SafetyStatus()[AOLD]* = 1 *OperationStatus()[XDSG]* = 1 DSG FET is disabled. Increment AOLDL counter
Latch	AOLDL counter ≥ ***AOLD:Latch Limit***	*SafetyAlert()[AOLDL]* = 0 *SafetyStatus()[AOLDL]* = 1 *OperationStatus()[XDSG]* = 1 DSG FET is disabled.
Recovery	*SafetyStatus()[AOLD]* = 1 for ***AOLD:Recovery*** time OR If ***Protection Configuration [LPEN]*** = 1 AND *AFEStatus()[LOAD_PRESENT]*= 0	*SafetyStatus()[AOLD]* = 0 *OperationStatus()[XDSG]* = 0 DSG FET returns to normal if *SafetyStatus[AOLDL]* = 0.
Latch Reset	*SafetyStatus()[AOLDL]* = 1 for A***OLD:Reset*** time	*SafetyStatus()[AOLDL]* = 0 Reset AOLDL counter *OperationStatus()[XDSG]* = 0 DSG FET returns to normal if *SafetyStatus()[AOLD]* = 0.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	AOLD	Threshold and Delay	H1	0x00	0xFF	0x00	—
Protections	AOLD	Latch Limit	U1	0	255	0	counts
Protections	AOLD	Counter Dec Delay	U1	0	255	10	s
Protections	AOLD	Recovery	U1	0	255	5	s
Protections	AOLD	Reset	U1	0	255	15	s

This register is representative of the BQ769x0 PROTECT 2 register.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	AOLD	Threshold and Delay	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
RSVD	OCD_D2	OCD_D1	OCD_D0	OCD_T3	OCD_T2	OCD_T1	OCD_T0

RSVD (Bit 7): Reserved. Do not use.

OCD_D2:0 (Bits 6–4): OCD Thresholds Delay Time

000 = 8 ms
001 = 20 ms
010 = 40 ms
011 = 80 ms
100 = 160 ms
101 = 320 ms
110 = 640 ms
111 = 1280 ms

OCD_T3:0 (Bits 3–0): OCD Thresholds with RSNS = 1

0000 = 17 mv
0001 = 22 mv
0010 = 28 mv
0011 = 33 mv
0100 = 39 mv
0101 = 44 mv
0110 = 50 mv
0111 = 56 mv
1000 = 61 mv
1001 = 67 mv
1010 = 72 mv
1011 = 78 mv
1100 = 83 mv
1101 = 89 mv
1110 = 94 mv
1111 = 100 mv

OCD_T3:0 (Bits 3–0): OCD Thresholds with RSNS = 0

0000 = 8 mv
0001 = 11 mv

0010 = 14 mv
0011 = 17 mv
0100 = 19 mv
0101 = 25 mv
0110 = 28 mv
0111 = 31 mv
1000 = 31 mv
1001 = 33 mv
1010 = 36 mv
1011 = 39 mv
1100 = 42 mv
1101 = 44 mv
1110 = 47 mv
1111 = 50 mv

3.6.2 Short Circuit in Discharge Protection

The device has a hardware-based short circuit in discharge protection with adjustable current and delay. See Section 3.1.4 for additional FET action options.

Status	Condition	Action
Normal	*Current()* > (***ASCD Threshold and Delay[2:0]***/R_{SENSE})	*SafetyAlert()[ASCDL]* = 0, if ASCDL counter = 0
Alert	ASCDL counter > 0	*SafetyAlert()[ASCDL]* = 1 Decrement ASCDL counter by one after each ***SCD:Counter Dec Delay*** period
Trip	*Current()* continuous ≤ (***ASCD Threshold and Delay[2:0]***/R_{SENSE}) for ***ASCD Threshold and Delay[7:4]*** duration	*SafetyStatus()[ASCD]* = 1 *OperationStatus()[XDSG]* = 1 DSG FET is disabled. Increment ASCDL counter
Latch	ASCD counter ≥ ***ASCD:Latch Limit***	*SafetyStatus()[ASCD]* = 0 *SafetyStatus()[ASCDL]* = 1 *OperationStatus()[XDSG]* = 1 DSG FET is disabled.
Recovery	*SafetyStatus()[ASCD]* = 1 for ***ASCD:Recovery*** time OR If ***Protection Configuration [LPEN]*** = 1 AND *AFEStatus()[LOAD_PRESENT]* = 0	*SafetyStatus()[ASCD]* = 0 *OperationStatus()[XDSG]* = 0 DSG FET returns to normal if *SafetyStatus()[ASCDL]* = 0.
Latch Reset	*SafetyStatus()[ASCDL]* = 1 for ***ASCD:Reset*** time	*SafetyStatus()[ASCDL]* = 0 *OperationStatus()[XDSG]* = 0 DSG FET returns to normal if *SafetyStatus()[ASCD]* = 0.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	ASCD	Threshold and Delay	H1	0x00	0xFF	0x00	—
Protections	ASCD	Latch Limit	U1	0	255	0	counts
Protections	ASCD	Counter Dec Delay	U1	0	255	10	s
Protections	ASCD	Recovery	U1	0	255	5	s
Protections	ASCD	Reset	U1	0	255	15	s

This register is representative of the BQ769x0 PROTECT 1 register.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	ASCD	Threshold and Delay	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
RSNS	RSVD	RSVD	SCD_D1	SCD_D0	SCD_T2	SCD_T1	SCD_T0

LEGEND: RSVD = Reserved Location

RSNS (Bit 7): AOLD and ASCD Thresholds Divisor

0 = 0.5 × AFE Protection Thresholds (default)

1 = Normal AFE Protection Thresholds

RSVD (Bits 6–5): Reserved. Do not use.

SCD_D1:0 (Bits 4–3): ASCD Delay Time

00 = 70 µs

01 = 100 µs

10 = 200 µs

11 = 400 µs

SCD_T2:0 (Bits 2–0): ASCD Thresholds with RSNS = 1

000 = 44 mv

001 = 67 mv

010 = 89 mv

011 = 111 mv

100 = 133 mv

101 = 155 mv

110 = 178 mv

111 = 200 mv

SCD_T2:0 (Bits 2–0): ASCD Thresholds with RSNS = 0

000 = 22 mv

001 = 33 mv

010 = 44 mv

011 = 56 mv

100 = 67 mv

101 = 78 mv

110 = 89 mv

111 = 100 mv

3.6.3 AFE ALERT OVRD Protection

The device can detect an external override signal sent to the companion BQ769x0 AFE that can cause permanent failure of the battery. This new option provides a temporary fault detection that acts on the FETs. The permanent failure option is not affected by this change.

Status	Condition	Action
Normal	*AFESysStat() [OVRD_ALERT]* = 0	*SafetyAlert()[AFE_OVRD]* = 0
Alert	*AFESysStat() [OVRD_ALERT]* = 1	*SafetyAlert()[AFE_OVRD]* = 1
Trip	*AFESysStat() [OVRD_ALERT]* = 1 for ***AFE External Override Delay*** duration	*SafetyAlert()[AFE_OVRD]* = 0 *SafetyStatus()[AFE_OVRD]* = 1 *OperationStatus()[XCHG, XDSG]* = 1 All FETs turn OFF
Recovery	*AFESysStat() [OVRD_ALERT]* = 0 for AFE External Override Recovery duration	*SafetyAlert()[AFE_OVRD]* = 0 *SafetyStatus()[AFE_OVRD]* = 0 *OperationStatus()[XCHG, XDSG]* = 0 CHG and DSG FETs allowed to turn ON

3.7 Temperature Protections

The device provides overtemperature and undertemperature protections based on cell temperature measurements. The cell temperature based protections are further divided into a protection-in-charging direction and discharging directions. This section describes in detail each of the protection functions.

For temperature reporting, the device supports a maximum of either three external thermistors or three internal temperature sensors. The selection of Internal or External temperature sensors is set by ***Settings:Temperature Enable[SOURCE]***. Unused temperature sensors should be disabled by clearing the corresponding flag in ***Settings:Temperature Enable[TS3][TS2][TS1]***.

The *Temperature()* command returns the cell temperature measurement. The MAC and extended command *DAStatus2()* also returns the temperature measurement from the enabled temperature sensors and the cell temperature.

The cell temperature based overtemperature and undertemperature safety provide protections in charge and discharge conditions. The battery pack is considered in CHARGE mode when *Battery[DSG]* = 0, where *Current()* > ***Chg Current Threshold***. The overtemperature and undertemperature in charging protections are active in this mode. The *Battery[DSG]* is set to 1 in a NON-CHARGE mode condition, which includes RELAX and DISCHARGE modes. The overtemperature and undertemperature in discharge protections are active in these two modes.

3.7.1 Overtemperature in Charge Protection

The device has an overtemperature protection for cells under charge.

Status	Condition	Action
Normal	Cell temperature in *Temperatures()* < ***OTC:Threshold*** OR not charging	*SafetyAlert()[OTC]* = 0
Alert	Cell temperature in *Temperatures()* ≥ ***OTC:Threshold*** AND charging	*SafetyAlert()[OTC]* = 1 *BatteryStatus()[TCA]* = 1
Trip	Cell temperature in *Temperatures()* ≥ ***OTC:Threshold*** AND charging for ***OTC:Delay*** duration	*SafetyAlert()[OTC]* = 0 *SafetyStatus()[OTC]* = 1 *BatteryStatus()[OTA]* = 1 *BatteryStatus()[TCA]* = 0 *OperationStatus()[XCHG]* = 1 Charging disabled if ***FET Options[OTFET]*** = 1
Recovery	*SafetyStatus()[OTC]* AND Cell temperature in *Temperatures()* ≤ ***OTC:Recovery***	*SafetyStatus()[OTC]* = 0 *BatteryStatus()[OTA]* = 0 *BatteryStatus()[TCA]* = 0 *OperationStatus()[XCHG]* = 0 Charging is allowed if ***FET Options[OTFET]*** = 1.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	OTC	Threshold	I2	–400	1500	550	0.1°C
Protections	OTC	Delay	U1	0	255	2	s
Protections	OTC	Recovery	I2	–400	1500	500	0.1°C

3.7.2 Overtemperature in Discharge Protection

The device has an overtemperature protection for cells in DISCHARGE state (that is, non-charging state with *BatteryStatus[DSG]* = 1).

Status	Condition	Action
Normal	Cell temperature in *Temperatures()* < ***OTD:Threshold*** OR charging	*SafetyAlert()[OTD]* = 0
Alert	Cell temperature in *Temperatures()* ≥ ***OTD:Threshold*** AND not charging (that is, *BatteryStatus[DSG]* = 1)	*SafetyAlert()[OTD]* = 1
Trip	Cell temperature in *Temperatures()* ≥ ***OTD:Threshold*** AND not charging (that is, *BatteryStatus[DSG]* = 1) for ***OTD:Delay*** duration	*SafetyAlert()[OTD]* = 0 *SafetyStatus()[OTD]* = 1 *BatteryStatus()[OTA]* = 1 Discharging is disabled AND *OperationStatus()[XDSG]* = 1 if ***FET Options[OTFET]*** = 1.
Recovery	*SafetyStatus()[OTD]* AND cell temperature in *Temperatures()* ≤ ***OTD:Recovery***	*SafetyStatus()[OTD]* = 0 *BatteryStatus()[OTA]* = 0 Discharging is allowed AND *OperationStatus()[XDSG]* = 0 if ***FET Options[OTFET]*** = 1.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	OTD	Threshold	I2	–400	1500	600	0.1°C
Protections	OTD	Delay	U1	0	255	2	s
Protections	OTD	Recovery	I2	–400	1500	550	0.1°C

3.7.3 Undertemperature in Charge Protection

The device has an undertemperature protection for cells in charge direction (that is, with *BatteryStatus[DSG]* = 0).

Status	Condition	Action
Normal	*Temperature()* > ***UTC:Threshold*** OR not charging	*SafetyAlert()[UTC]* = 0
Alert	*Temperature()* ≤ ***UTC:Threshold*** AND charging	*SafetyAlert()[UTC]* = 1
Trip	*Temperature()* ≤ ***UTC:Threshold*** AND Charging for ***UTC:Delay*** duration	*SafetyAlert()[UTC]* = 0 *SafetyStatus()[UTC]* = 1 *OperationStatus()[XCHG]* = 1
Recovery	*SafetyStatus()[UTC]* AND *Temperature()* ≥ ***UTC:Recovery***	*SafetyStatus()[UTC]* = 0 *OperationStatus()[XCHG]* = 0

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	UTC	Threshold	I2	–400	1500	0	0.1°C
Protections	UTC	Delay	U1	0	255	2	s
Protections	UTC	Recovery	I2	–400	1500	50	0.1°C

3.7.4 Undertemperature in Discharge Protection

The device has an undertemperature protection for cells in DISCHARGE state (that is, non-charging state with *BatteryStatus[DSG]* = 1).

Status	Condition	Action
Normal	*Temperature()* > ***UTD:Threshold*** OR charging	*SafetyAlert()[UTD]* = 0
Alert	*Temperature()* ≤ ***UTD:Threshold*** AND Not charging (that is, *BatteryStatus[DSG]* = 1)	*SafetyAlert()[UTD]* = 1
Trip	*Temperature()* ≤ ***UTD:Threshold*** AND Not charging (that is, *BatteryStatus[DSG]* = 1) for ***UTD:Delay*** duration	*SafetyAlert()[UTD]* = 0 *SafetyStatus()[UTD]* = 1 *OperationStatus()[XDSG]* = 1

Status	Condition	Action
Recovery	*SafetyStatus()[UTD]* AND *Temperature()* ≥ ***UTD:Recovery***	*SafetyStatus()[UTD]* = 0 *OperationStatus()[XDSG]* = 0

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	UTD	Threshold	I2	–400	1500	0	0.1°C
Protections	UTD	Delay	U1	0	255	2	s
Protections	UTD	Recovery	I2	–400	1500	50	0.1°C

3.7.5 Overtemperature FET Protection

The device has an overtemperature protection to limit the FET temperature.

NOTE: This feature is only available when using the BQ76930x or BQ76940x device: There is only one external TS available in the BQ76920 and this is used for cell temperature.

Status	Condition	Action
Normal	*FETTemperature()* < *OTF:Threshold*	*SafetyAlert()[OTF]* = 0
Alert	*FETTemperature()* ≥ *OTF:Threshold*	*SafetyAlert()[OTF]* = 1 *BatteryStatus()[TDA]* = 1, *[TCA]* = 1
Trip	*FETTemperature()* ≥ *OTF:Threshold* for *OTF:Delay* duration	*SafetyAlert()[OTF]* = 0 *SafetyStatus()[OTF]* = 1 *BatteryStatus()[OTA]* = 1 *BatteryStatus()[TDA]* = 0, [TCA] = 0 *OperationStatus()[XCHG][XDSG]* = 1,1 if ***FET Options[OTFET]*** = 1
Recovery	*SafetyStatus()[OTF]* AND *FETTemperature()* ≤ *OTF:Recovery*	*SafetyStatus()[OTF]* = 0 *BatteryStatus()[OTA]* = 0 *BatteryStatus()[TDA]* = 0, *[TCA]* = 0 *OperationStatus()[XCHG][XDSG]* = 0,0

3.8 Precharge Timeout Protection

The device can measure the precharge time and stop charging if it exceeds the adjustable period.

Status	Condition	Action
Enable	*Current()* > ***PTO:Charge Threshold*** AND *ChargingStatus()[PCHG]* = 1	Start PTO timer *SafetyAlert()[PTOS]* = 0
Suspend or Recovery	*Current()* < ***PTO:Suspend Threshold***	Stop PTO timer *SafetyAlert()[PTOS]* = 1
Trip	PTO timer > ***PTO:Delay***	Stop PTO timer *SafetyStatus()[PTO]* = 1 *BatteryStatus()[TCA]* = 1 Charging is not allowed.
Reset	*SafetyStatus()[PTO]* = 1 AND (Discharge by an amount of ***PTO:Reset***)	Stop and reset PTO timer *SafetyAlert()[PTOS]* = 0 *SafetyStatus()[PTO]* = 0 *BatteryStatus()[TCA]* = 0 Charging is allowed.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	PTO	Charge Threshold	I2	–32768	32767	2000	mA
Protections	PTO	Suspend Threshold	I2	–32768	32767	1800	mA
Protections	PTO	Delay	U2	0	65535	1800	s
Protections	PTO	Reset	I2	–32768	32767	2	mAh

3.9 Fast Charge Timeout Protection

The device can measure the charge time and stop charging if it exceeds the adjustable period.

Status	Condition	Action
Enable	*Current()* > CTO:Charge Threshold	Start CTO timer *SafetyAlert()[CTOS]* = 0
Suspend or Recovery	*Current()* < CTO:Suspend Threshold	Stop CTO timer *SafetyAlert()[CTOS]* = 1
Trip	CTO time > CTO:Delay	Stop CTO timer *SafetyStatus()[CTO]* = 1 *BatteryStatus()[TCA]* = 1 Charging is not allowed.
Reset	*SafetyStatus()[CTO]* = 1 AND (Discharge by an amount of ***CTO:Reset***)	Stop and reset CTO timer *SafetyAlert()[CTOS]* = 0 *SafetyStatus()[CTO]* = 0 *BatteryStatus()[TCA]* = 0 Charging is allowed.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	CTO	Charge Threshold	I2	–32768	32767	2500	mA
Protections	CTO	Suspend Threshold	I2	–32768	32767	2000	mA
Protections	CTO	Delay	U2	0	65535	54000	s
Protections	CTO	Reset	I2	–32768	32767	2	mAh

3.10 Overcharge Protection

The device can prevent continuing charging if the pack is charged in excess over *FullChargeCapacity()*. While *RemainingCapacity()* never reports a value higher than *FullChargeCapacity()* in the device registers, it is tracked to higher values internally to protect against overcharging.

Status	Condition	Action
Normal	*RemainingCapacity()* < *FullChargeCapacity()*	*SafetyAlert()[OC]* = 0
Alert	*RemainingCapacity()* ≥ *FullChargeCapacity()*	*SafetyAlert()[OC]* = 1
Trip	*RemainingCapacity()* ≥ *FullChargeCapacity()* + ***OC:Threshold***	*SafetyAlert()[OC]* = 0 *SafetyStatus()[OC]* = 1 *BatteryStatus()[TCA]* = 1, *[OCA]* = 1 if the device is in CHARGE state (that is, *BatteryStatus[DSG]* = 0). *OperationStatus()[XCHG]* = 1 Charging is not allowed.
Recovery	*SafetyStatus()[OC]* = 1 AND continuous discharge of ***Recovery*** OR *RemainingStateOfCharge()* < ***OC:RSOC Recovery***	*SafetyStatus()[OC]* = 0 *BatteryStatus()[TCA]* = 0, *[OCA]* = 0 *OperationStatus()[XCHG]* = 0 Charging is allowed.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Protections	OC	Threshold	I2	–32768	32767	300	mAh
Protections	OC	Recovery	I2	–32768	32767	2	mAh
Protections	OC	RSOC Recovery	U1	0%	100%	90%	

Permanent Fail

4.1 Introduction

The device can permanently disable the use of the battery pack in case of a severe failure. The permanent failure checks, except for IFC and DFW, can be individually enabled or disabled by setting the appropriate bit in ***Settings:Enabled PF A***, ***Settings:Enabled PF B***, ***Settings:Enabled PF C***, and ***Settings:Enabled PF D***. All permanent failure checks except for IFC and DFW are disabled until *ManufacturingStatus()[PF]* is set. When any *PFStatus()* bit is set, the device enters PERMANENT FAIL mode and the following actions are taken in sequence:

1. Precharge, charge, and discharge FETs are turned off.
2. *OperationStatus()[PF]* = 1
3. The following SBS data is changed: *BatteryStatus()[TCA]* = 1, *BatteryStatus()[TDA]* = 1, *ChargingCurrent()* = 0, and *ChargingVoltage()* = 0.
4. A backup of the internal AFE hardware registers are written to data flash: ***AFE Status***, ***AFE Config***, ***AFE VCx***, and ***AFE Data***.
5. The black box data of the last three *SafetyStatus()* changes leading up to PF with the time difference is written into the black box data flash along with the 1st *PFStatus()* value.
6. The following SBS values are preserved in data flash for failure analysis:
   - *SafetyAlert()*
   - *SafetyStatus()*
   - *PFAlert()*
   - *PFStatus()*
   - *OperationStatus()*
   - *ChargingStatus()*
   - *GaugingStatus()*
   - Voltages in *DAStatus1()*
   - *Current()*
   - TS1, TS2, and TS3 from *DAStatus2()*
7. Data flash writing is disabled (except to store subsequent *PFStatus()* flags).
8. The SAFE pin is driven high if configured for specific failures and *Voltage()* is above 3500 mV or there is a CHG FET (CFETF) or DSG FET (DFETF) failure. The SAFE pin will remain asserted until the ***Fuse Blow Timeout*** expired.

While the device is in PERMANENT FAIL mode, any new *SafetyAlert()*, *SafetyStatus()*, *PFAlert()*, and *PFStatus()* flags that are set are added to the permanent fail log. Any new *PFStatus()* flags that occur during PERMANENT FAIL mode can trigger the SAFE pin. In addition, new *PFStatus()* flags are recorded in the Black Box Recorder 2nd and 3rd PF Status entries.

4.2 Permanent Failure Configuration

The following configuration registers allow the various permanent failure detection features to be enabled or disabled. If disabled (default), the feature takes no action including setting flags in *PFAlert()* or *PFStatus()*.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Permanent Failure	Enabled PF A	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
DFETF	CFETF	VIMR	SOT	SOCD	SOCC	SOV	SUV

DFETF (Bit 7): Discharge FET
- 1 = Enabled
- 0 = Disabled (default)

CFETF (Bit 6): Charge FET
- 1 = Enabled
- 0 = Disabled (default)

VIMR (Bit 5): Voltage imbalance at rest
- 1 = Enabled
- 0 = Disabled (default)

SOT (Bit 4): Safety Overtemperature Cell
- 1 = Enabled
- 0 = Disabled (default)

SOCD (Bit 3): Safety Overcurrent in Discharge
- 1 = Enabled
- 0 = Disabled (default)

SOCC (Bit 2): Safety Overcurrent in Charge
- 1 = Enabled
- 0 = Disabled (default)

SOV (Bit 1): Safety overvoltage
- 1 = Enabled
- 0 = Disabled (default). This feature cannot be stopped from turning the appropriate FETs OFF as this is a hardware feature of the companion AFE.

SUV (Bit 0): Safety undervoltage
- 1 = Enabled
- 0 = Disabled (default). This feature cannot be stopped from turning the appropriate FETs OFF as this is a hardware feature of the companion AFE.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Permanent Failure	Enabled PF B	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
SOTF	TS3	TS2	TS1	AFE_XRDY	AFE_OVRD	AFEC	AFER

SOTF (Bit 7): Safety Overtemperature FET
- 1 = Enabled
- 0 = Disabled (default)

TS3 (Bit 6): Temperature sensor 3
- 1 = Enabled
- 0 = Disabled (default)

TS2 (Bit 5): Temperature sensor 2
- 1 = Enabled
- 0 = Disabled (default)

TS1 (Bit 4): Temperature sensor 1
- 1 = Enabled
- 0 = Disabled (default)

AFE_XRDY (Bit 3): Companion AFE XREADY
- 1 = Enabled
- 0 = Disabled (default)

AFE_OVRD (Bit 2): Companion AFE OVERRIDE
- 1 = Enabled
- 0 = Disabled (default)

AFEC (Bit 1): AFE Communication
- 1 = Enabled
- 0 = Disabled (default)

AFER (Bit 0): AFE Register
- 1 = Enabled
- 0 = Disabled (default)

4.3 Enabling Use of the SAFE Pin

The SAFE pin can be enabled or disabled for use for any of the enabled protections through the settings in the following:

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Fuse	PF SAFE A	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
DFETF	CFETF	VIMR	SOT	SOCD	SOCC	SOV	SUV

DFETF (Bit 7): Discharge FET
- 1 = Enabled
- 0 = Disabled (default)

CFETF (Bit 6): Charge FET

1 = Enabled
0 = Disabled (default)

VIMR (Bit 5): Voltage Imbalance at Rest
1 = Enabled
0 = Disabled (default)

SOT (Bit 4): Safety Overtemperature Cell
1 = Enabled
0 = Disabled (default)

SOCD (Bit 3): Safety Overcurrent in Discharge
1 = Enabled
0 = Disabled (default)

SOCC (Bit 2): Safety Overcurrent in Charge
1 = Enabled
0 = Disabled (default)

SOV (Bit 1): Safety Overvoltage
1 = Enabled
0 = Disabled (default)

SUV (Bit 0): Safety Undervoltage
1 = Enabled
0 = Disabled (default)

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Fuse	PF SAFE B	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
SOTF	TS3	TS2	TS1	AFE_XRDY	AFE_OVRD	AFEC	AFER

SOTF (Bit 7): Safety Overtemperature FET
1 = Enabled
0 = Disabled (default)

TS3 (Bit 6): TS3
1 = Enabled
0 = Disabled (default)

TS2 (Bit 5): TS2
1 = Enabled
0 = Disabled (default)

TS1 (Bit 4): TS1
1 = Enabled
0 = Disabled (default)

AFE_XRDY (Bit 3): AFE XREADY
1 = Enabled
0 = Disabled (default)

AFE_OVRD (Bit 2): AFE Override
- 1 = Enabled
- 0 = Disabled (default)

AFEC (Bit 1): AFE Communication
- 1 = Enabled
- 0 = Disabled (default)

AFER (Bit 0): AFE Register
- 1 = Enabled
- 0 = Disabled (default)

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Fuse	PF SAFE C	Hex	1	0x00	0xFF	0x00	—

7	6	5	4	3	2	1	0
RSVD	RSVD	RSVD	RSVD	RSVD	RSVD	DFW	IFC

RSVD (Bits 7–2): Reserved

DFW (Bit 1): Data flash write
- 1 = Enabled
- 0 = Disabled (default)

IFC (Bit 0): Instruction flash checksum
- 1 = Enabled
- 0 = Disabled (default)

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
PF Status	Device Status Data	Fuse Flag	Hex	2	0x0	0xFFFF	0x0	Hex

Fuse Flag is set to 0x0 by default, but will be set to 0x3672 to indicate a fuse blow.

The BQ78350-R1A has a minimum voltage required to attempt to blow a fuse through SAFE activation. This is a pack-based value of 3500 mV. Voltage scaling (VSCALE) should be enabled if the supported battery pack voltage is higher than 32767 mV. This value is automatically internally adjusted for any VSCALE setting. FET failures bypass this requirement to activate SAFE.

4.4 Safety Cell Undervoltage Permanent Fail

The BQ78350-R1A uses the UV Protection function of the companion AFE for this feature and can be configured to permanently disable the battery in the case of severe undervoltage in any of the cells. This feature cannot be disabled.

The voltage threshold setting is set in ***AFE SUV:Threshold***, which the device will map to the available settings in the companion AFE with the maximum setting of 3131 mV and the minimum of 1568 mV.

The delay timing configuration for this feature is combined in the same register with the delay time of the Safety Overvoltage feature.

Status	Condition	Action
Normal	*AFEStatus() [UV]* = 0	*PFStatus()[SUV]* = 0
Trip	*AFEStatus() [UV]* = 1	*PFStatus()[SUV]* = 1 *BatteryStatus()[FD]* = 1 *BatteryStatus()[TDA]* = 1 *BatteryStatus()[TCA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	AFE SUV	Threshold	I2	1580	3100	1750	mV
Permanent Fail	AFE SOV/AFE SUV	SOV and SUV Delay	U1	0	255	2	s

This register is representative of the BQ769x0 PROTECT 3 register.

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	AFE SOV/AFE SUV	SOV and SUV Delay	Hex	1	0x00	0xF0	0x50	—

7	6	5	4	3	2	1	0
SUV_D1	SUV_D0	SOV_D1	SOV_D0	RSVD	RSVD	RSVD	RSVD

LEGEND: R/W = Read/Write; R = Read only; -*n* = value after reset

SUV_D1:0 (Bits 7–6): Safety Undervoltage Delay Time

00 = 1 s
01 = 4 s
10 = 8 s
11 = 16 s

SOV_D1:0 (Bits 5–4): Safety Overvoltage Delay Time

00 = 1 s
01 = 2 s
10 = 4 s
11 = 8 s

RSVD: (Bits 3–0): Reserved

4.5 Safety Cell Overvoltage Permanent Fail

The BQ78350-R1A uses the OV Protection function of the companion AFE for this feature and can be configured to permanently disable the battery in the case of severe overvoltage in any of the cells. This feature cannot be disabled.

The voltage threshold setting is set in ***AFE SOV:Threshold***, which the device will map to the available settings in the companion AFE with the maximum setting of 4703 mV and the minimum of 3140 mV.

The delay timing configuration for this feature is combined in the same register with the delay time of the Safety Undervoltage feature.

Status	Condition	Action
Normal	*AFEStatus() [OV]* = 0	*PFStatus()[SOV]* = 0
Trip	*AFEStatus() [OV]* = 1	*PFStatus()[SOV]* = 1 *BatteryStatus()[TDA]* = 1 *BatteryStatus()[TCA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	AFE SOV	Threshold	I2	3150	4700	4350	mV

4.6 Safety Overcurrent in Charge Permanent Fail

The device can permanently disable the battery in the case of a severe OVERCURRENT IN CHARGE state.

Status	Condition	Action
Normal	*Current()* < ***SOCC:Threshold***	*PFAlert()[SOCC]* = 0
Alert	*Current()* ≥ ***SOCC:Threshold***	*PFAlert()[SOCC]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[OCA]* = 1
Trip	*Current()* ≥ ***SOCC:Threshold*** for ***SOCC:Delay*** duration	*PFAlert()[SOCC]* = 0 *PFStatus()[SOCC]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1 *BatteryStatus()[OCA]* = 1

4.7 Safety Overcurrent in Discharge Permanent Fail

The device can permanently disable the battery in the case of severe overcurrent in DISCHARGE or RELAX state.

Status	Condition	Action
Normal	*Current()* > ***SOCD:Threshold***	*PFAlert()[SOCD]* = 0
Alert	*Current()* ≤ ***SOCD:Threshold***	*PFAlert()[SOCD]* = 1 *BatteryStatus()[TDA]* = 1
Trip	*Current()* ≤ ***SOCD:Threshold*** for ***SOCD:Delay*** duration	*PFAlert()[SOCD]* = 0 *PFStatus()[SOCD]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

4.8 Safety Overtemperature Cell Permanent Fail

The device can permanently disable the battery pack in case of severe overtemperature of the cells detected using the external TS1...3 temperature sensor(s), which are configured to report *Temperature()*. The *Temperature()* measurement configuration is controlled by setting the corresponding flag in ***DA Configuration***.

Status	Condition	Action
Normal	Cell temperature in *DAStatus2()* < ***SOT:Threshold***	*PFAlert()[SOT]* = 0
Alert	Cell temperature in *DAStatus2()* ≥ ***SOT:Threshold***	*PFAlert()[SOT]* = 1 *BatteryStatus()[OTA]* = 1
Trip	Cell temperature in *DAStatus2()* continuous ≥ ***SOT:Threshold*** for ***SOT:Delay*** duration	*PFAlert()[SOT]* = 0 *PFStatus()[SOT]* = 1 *BatteryStatus()[OTA]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	SOT	Threshold	I2	–400	1500	650	0.1°C
Permanent Fail	SOT	Delay	U1	0	255	5	s

4.9 Safety Overtemperature FET (SOTF) Permanent Fail

The device can disable the battery pack permanently in case of severe overtemperature on the power FET. The temperature sensor(s) can be configured to report as FET Temperature in *DAStatus2()* by setting the corresponding flags in ***Temperature Mode*** and ***DA Configuration[FTEMP]***.

Status	Condition	Action
Normal	FET Temperature in *DAStatus2()* < ***SOTF:Threshold***	*PFAlert()[SOTF]* = 0
Alert	FET Temperature in *DAStatus2()* ≥ ***SOTF:Threshold***	*PFAlert()[SOTF]* = 1 *BatteryStatus()[OTA]* = 1
Trip	FET Temperature in *DAStatus2()* continuous ≥ ***SOTF:Threshold for SOTF:Delay duration***	*PFAlert()[SOTF]* = 0 *PFStatus()[SOTF]* = 1 *BatteryStatus()[OTA]* = 1

4.10 Voltage Imbalance at Rest Permanent Fail

The device can permanently disable the battery pack in case of a voltage difference between the cells in a stack while at rest.

Status	Condition	Action
Normal	*CellVoltage1..15()* < ***VIMR:Check Voltage*** OR |*Current()*| > ***VIMR:Check Current*** OR Δ(*CellVoltage1..15()*) < ***VIMR:Delta Threshold***	*PFAlert()[VIMR]* = 0
Alert	Any(*CellVoltage1..15()*) ≥ ***VIMR:Check Voltage*** AND |*Current()*| < ***VIMR:Check Current*** for ***VIMR:Duration*** AND Δ(*CellVoltage1..15()*) ≥ ***VIMR:Delta Threshold***	*PFAlert()[VIMR]* = 1
Trip	Any(*CellVoltage1..15()*) ≥ ***VIMR:Check Voltage*** AND |*Current()*| < ***VIMR:Check Current*** for ***VIMR:Duration*** AND Δ(*CellVoltage1..15()*) ≥ ***VIMR:Delta Threshold*** for ***VIMR:Delta Delay***	*PFAlert()[VIMR]* = 0 *PFStatus()[VIMR]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	VIMR	Check Voltage	I2	0	5000	5000	mV
Permanent Fail	VIMR	Check Current	I2	0	32767	10	mA
Permanent Fail	VIMR	Delta Threshold	I2	0	5000	500	mV
Permanent Fail	VIMR	Delta Delay	U1	0	255	5	s
Permanent Fail	VIMR	Duration	U2	0	65535	100	s

4.11 Charge FET Permanent Fail

The device can permanently disable the battery pack in case the charge FET is not working properly.

Status	Condition	Action
Normal	CHG FET off AND *Current()* < ***CFET:OFF Threshold***	*PFAlert()[CFETF]* = 0
Alert	CHG FET off AND *Current()* ≥ ***CFET:OFF Threshold***	*PFAlert()[CFETF]* = 1
Trip	CHG FET off AND *Current()* continuously ≥ ***CFET:OFF Threshold*** for ***CFET:OFF Delay*** duration	*PFAlert()[CFETF]* = 0 *PFStatus()[CFETF]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	CFETF	OFF Threshold	I2	0	500	5	mA
Permanent Fail	CFETF	Delay	U1	0	255	5	s

4.12 Discharge FET Permanent Fail

The device can permanently disable the battery pack in case the discharge FET is not working properly.

Status	Condition	Action
Normal	DSG FET off AND *Current()* > ***DFET:OFF Threshold***	*PFAlert()[DFETF]* = 0
Alert	DSG FET off AND *Current()* ≤ ***DFET:OFF Threshold***	*PFAlert()[DFETF]* = 1
Trip	DSG FET off AND *Current()* continuously ≤ ***DFET:OFF Threshold*** for ***DFET:OFF Delay*** duration	*PFAlert()[DFETF]* = 0 *PFStatus()[DFETF]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	DFET	OFF Threshold	I2	–500	0	–5	mA
Permanent Fail	DFET	Delay	U1	0	255	5	s

4.13 External Override Permanent Fail

The device can detect an external override signal sent to the companion BQ769x0 AFE, which can cause permanent failure of the battery. This can be used to indicate to the BQ78350-R1A that an external circuit, such as an independent voltage protection circuit, has disabled the battery permanently.

Status	Condition	Action
Normal	*AFESysStat() [OVRD_ALERT]* = 0	*PFAlert()[AFE_OVRD]* = 0
Alert	*AFESysStat() [OVRD_ALERT]* = 1	*PFAlert()[AFE_OVRD]* = 1
Trip	*AFESysStat() [OVRD_ALERT]* = 1 continuously for ***AFE External Override: Delay*** duration	*PFAlert()[AFE_OVRD]* = 0 *PFStatus()[AFE_OVRD]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	AFE External Override	Delay	U1	0	255	5	s

4.14 AFE Register Permanent Fail

The device compares the AFE hardware register periodically with a RAM backup and corrects any errors. If any errors are found during the check, the device increments the AFE register fail counter. If the comparison fails too many times, the device disables the pack permanently.

Status	Condition	Action
Normal	AFE register fail counter = 0	*PFAlert()[AFER]* = 0 Compare AFE register and RAM backup every ***AFER:Compare Period***
Alert	AFE register fail counter > 0	*PFAlert()[AFER]* = 1 Decrement AFE register fail counter by one after each ***AFER:Delay Period*** Compare AFE register and RAM backup every ***AFER:Compare Period***
Trip	AFE register fail counter ≥ ***AFER:Threshold***	*PFAlert()[AFER]* = 0 *PFStatus()[AFER]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	AFER	Threshold	U1	0	255	100	counts
Permanent Fail	AFER	Delay Period	U1	0	255	2	s
Permanent Fail	AFER	Compare Period	U1	0	255	5	s

4.15 AFE Communication Permanent Fail

The device monitors the internal communication to the AFE hardware and increments the AFE read/write fail counter on any communication error. If the read or write fails exceed a limit within a configurable timeframe, the device disables the pack permanently.

Status	Condition	Action
Normal	AFE read/write fail counter = 0	*PFAlert()[AFEC]* = 0
Alert	AFE read/write fail counter > 0	*PFAlert()[AFEC]* = 1 Decrement AFE read/write fail counter by one after each ***AFEC:Delay Period***
Trip	Read and Write Fail counter ≥ ***AFEC:Threshold***	*PFAlert()[AFEC]* = 0 *PFStatus()[AFEC]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	AFEC	Threshold	U1	0	255	100	counts
Permanent Fail	AFEC	Delay Period	U1	0	255	5	s

4.16 AFE XREADY Permanent Fail

The companion BQ769x0 AFE includes an internal self-check, and if this check fails, then the XREADY bit is set. Each time the BQ78350-R1A reads the AFE it checks this bit, and if it is set, then increments an internal counter. If this counter reaches a configurable limit, then the device disables the pack permanently.

Status	Condition	Action
Normal	XREADY counter = 0	*PFAlert()[AFE_XRDY]* = 0
Alert	XREADY counter > 0	*PFAlert()[AFE_XRDY]* = 1 Decrement AFE_XRDY counter by one after each ***AFE XREADY:Delay*** period
Trip	XREADY counter ≥ ***XREADY: Threshold***	*PFAlert()[AFE_XRDY]* = 0 *PFStatus()[AFE_XRDY]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	AFE XREADY	Threshold	U1	0	255	100	counts
Permanent Fail	AFE XREADY	Delay Period	U1	0	255	5	s

4.17 Instruction Flash (IF) Checksum Permanent Fail

The device can permanently disable the battery if it detects a difference between the stored IF checksum and the calculated IF checksum only following a device reset.

Status	Condition	Action
Normal	Stored and calculated IF checksum match	—
Trip	Stored and calculated IF checksum after reset does not match.	*PFStatus()[IFC]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

4.18 Data Flash (DF) Permanent Fail

The device can permanently disable the battery in case a data flash write fails.

NOTE: A DF write failure causes the gauge to disable further DF writes.

Status	Condition	Action
Normal	Data flash write ok	—
Trip	Data flash write not successful	*PFStatus()[DFW]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

4.19 Open Thermistor Permanent Fail (TS1, TS2, TS3)

The device can permanently disable the battery if it detects an open thermistor on TS1, TS2, or TS3. This feature is only available when the BQ78350-R1A is used in conjunction with the BQ76930 or the BQ76940.

Status	Condition	Action
Normal	TS1 Temperature > ***Open Thermistor:Threshold***	*PFAlert()[TS1]* = 0
Normal	TS2 Temperature > ***Open Thermistor:Threshold***	*PFAlert()[TS2]* = 0
Normal	TS3 Temperature > ***Open Thermistor:Threshold***	*PFAlert()[TS3]* = 0
Alert	TS1 Temperature ≤ ***Open Thermistor:Threshold***	*PFAlert()[TS1]* = 1
Alert	TS2 Temperature ≤ ***Open Thermistor:Threshold***	*PFAlert()[TS2]* = 1
Alert	TS3 Temperature ≤ ***Open Thermistor:Threshold***	*PFAlert()[TS3]* = 1
Trip	TS1 Temperature ≤ ***Open Thermistor:Threshold*** for ***Open Thermistor:Delay*** duration	*PFAlert()[TS1]* = 0 *PFStatus()[TS1]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1
Trip	TS2 Temperature ≤ ***Open Thermistor:Threshold*** for ***Open Thermistor:Delay*** duration	*PFAlert()[TS2]* = 0 *PFStatus()[TS2]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1
Trip	TS3 Temperature ≤ ***Open Thermistor:Threshold*** for ***Open Thermistor:Delay*** duration	*PFAlert()[TS3]* = 0 *PFStatus()[TS3]* = 1 *BatteryStatus()[TCA]* = 1 *BatteryStatus()[TDA]* = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Permanent Fail	Open Thermistor	Threshold	I2	0	32767	2232	0.1 K
Permanent Fail	Open Thermistor	Delay	U1	0	255	5	s

4.20 PF Status Snapshot Data Flash

4.20.1 Device Status Data

Class	Subclass	Name	Type	Min	Max	Default	Description
PF Status	Device Status Data	Safety Alert A	H1	0x00	0xFF	0	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Status A	H1	0x00	0xFF	0	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Alert B	H1	0x00	0xFF	0	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Status B	H1	0x00	0xFF	0	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Alert C	H1	0x00	0xFF	0	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Status C	H1	0x00	0xFF	0	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Alert D	H1	0x00	0xFF	0	Accumulated safety flags since PF event
PF Status	Device Status Data	Safety Status D	H1	0x00	0xFF	0	Accumulated safety flags since PF event

Class	Subclass	Name	Type	Min	Max	Default	Description
PF Status	Device Status Data	PF Alert A	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	PF Status A	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	PF Alert B	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	PF Status B	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	PF Alert C	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	PF Status C	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	PF Alert D	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	PF Status D	H1	0x00	0xFF	0	Accumulated PF flags since PF event
PF Status	Device Status Data	SAFE Flag	H2	0x0000	0xFFFF	0	Flag set to indicate SAFE activation
PF Status	Device Status Data	Operation Status A	H2	0x0000	0xFFFF	0	*OperationStatus()* data at the time of the PF event
PF Status	Device Status Data	Operation Status B	H2	0x0000	0xFFFF	0	*OperationStatus()* data at the time of the PF event
PF Status	Device Status Data	Charging Status A	H1	0x00	0xFF	0	*ChargingStatus()* data at the time of the PF event
PF Status	Device Status Data	Charging Status B	H1	0x00	0xFF	0	*ChargingStatus()* data at the time of the PF event
PF Status	Device Status Data	Gauging Status A	H1	0x00	0xFF	0	*GaugingStatus()* data at the time of the PF event
PF Status	Device Status Data	Gauging Status B	H2	0x0000	0xFFFF	0	*GaugingStatus()* data at the time of the PF event

4.20.2 Device Voltage Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Voltage Data	Cell Voltage 1	I2	0	32767	0	mV	Cell 1 voltage
PF Status	Device Voltage Data	Cell Voltage 2	I2	0	32767	0	mV	Cell 2 voltage
PF Status	Device Voltage Data	Cell Voltage 3	I2	0	32767	0	mV	Cell 3 voltage
PF Status	Device Voltage Data	Cell Voltage 4	I2	0	32767	0	mV	Cell 4 voltage
PF Status	Device Voltage Data	Cell Voltage 5	I2	0	32767	0	mV	Cell 5 voltage
PF Status	Device Voltage Data	Cell Voltage 6	I2	0	32767	0	mV	Cell 6 voltage
PF Status	Device Voltage Data	Cell Voltage 7	I2	0	32767	0	mV	Cell 7 voltage
PF Status	Device Voltage Data	Cell Voltage 8	I2	0	32767	0	mV	Cell 8 voltage
PF Status	Device Voltage Data	Cell Voltage 9	I2	0	32767	0	mV	Cell 9 voltage
PF Status	Device Voltage Data	Cell Voltage 10	I2	0	32767	0	mV	Cell 10 voltage
PF Status	Device Voltage Data	Cell Voltage 11	I2	0	32767	0	mV	Cell 11 voltage
PF Status	Device Voltage Data	Cell Voltage 12	I2	0	32767	0	mV	Cell 12 voltage
PF Status	Device Voltage Data	Cell Voltage 13	I2	0	32767	0	mV	Cell 13 voltage
PF Status	Device Voltage Data	Cell Voltage 14	I2	0	32767	0	mV	Cell 14 voltage

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Voltage Data	Cell Voltage 15	I2	0	32767	0	mV	Cell 15 voltage
PF Status	Device Voltage Data	Bat Direct Voltage	I2	0	32767	0	mV	Cell stack voltage

4.20.3 Device Current Data

Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Current Data	Current	I2	–32768	32767	0	mA	*Current()*

4.20.4 Device Temperature Data

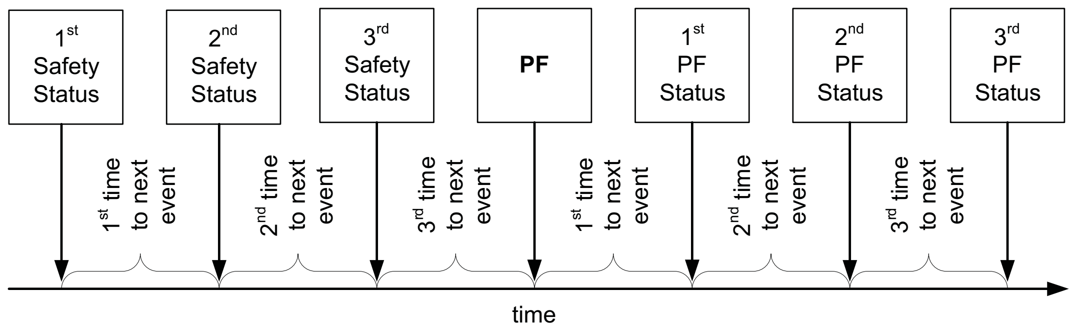
Class	Subclass	Name	Type	Min	Max	Default	Unit	Description
PF Status	Device Temperature Data	TS1 Temperature	I2	–32768	32767	0	0.1 K	TS1 temperature
PF Status	Device Temperature Data	TS2 Temperature	I2	–32768	32767	0	0.1 K	TS2 temperature
PF Status	Device Temperature Data	TS3 Temperature	I2	–32768	32767	0	0.1 K	TS3 temperature

4.20.5 AFE Regs

Class	Subclass	Name	Type	Length in Bytes	Min	Max	Default
PF Status	AFE Regs	AFE Sys Stat	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Cell Balance 1	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Cell Balance 2	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Cell Balance 3	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Sys Control 1	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Sys Control 2	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Protection 1	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Protection 2	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE Protection 3	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE OV Trip	H1	1	0x00	0xFF	0x00
PF Status	AFE Regs	AFE UV Trip	H1	1	0x00	0xFF	0x00

4.21 Black Box Recorder

The Black Box Recorder maintains the last three updates of *SafetyStatus()* in memory. When entering PERMANENT FAIL mode, this information is written to data flash in addition to the first three updates of *PFStatus()* after the PF event.



NOTE: This information is useful in failure analysis, and can provide a full recording of the events and conditions leading up to the permanent failure.

If there were less than three safety events before PF, then some information will be left blank.

4.21.1 Black Box Recorded Data

4.21.1.1 Safety Status

Class	Subclass	Name	Type	Min	Max	Default
Black Box	Safety Status	1st Safety Status 0–7	H1	0x00	0xFF	0
Black Box	Safety Status	1st Safety Status 8–15	H1	0x00	0xFF	0
Black Box	Safety Status	1st Safety Status 16–23	H1	0x00	0xFF	0
Black Box	Safety Status	1st Safety Status 24–31	H1	0x00	0xFF	0
Black Box	Safety Status	1st Time to Next Event	U1	0	255	0
Black Box	Safety Status	2nd Safety Status 0–7	H1	0x00	0xFF	0
Black Box	Safety Status	2nd Safety Status 8–15	H1	0x00	0xFF	0
Black Box	Safety Status	2nd Safety Status 16–23	H1	0x00	0xFF	0
Black Box	Safety Status	2nd Safety Status 24–31	H1	0x00	0xFF	0
Black Box	Safety Status	2nd Time to Next Event	U1	0	255	0
Black Box	Safety Status	3rd Safety Status 0–7	H1	0x00	0xFF	0
Black Box	Safety Status	3rd Safety Status 8–15	H1	0x00	0xFF	0
Black Box	Safety Status	3rd Safety Status 16–23	H1	0x00	0xFF	0
Black Box	Safety Status	3rd Safety Status 24–31	H1	0x00	0xFF	0
Black Box	Safety Status	3rd Time to Next Event	U1	0	255	0

4.21.1.2 PF Status

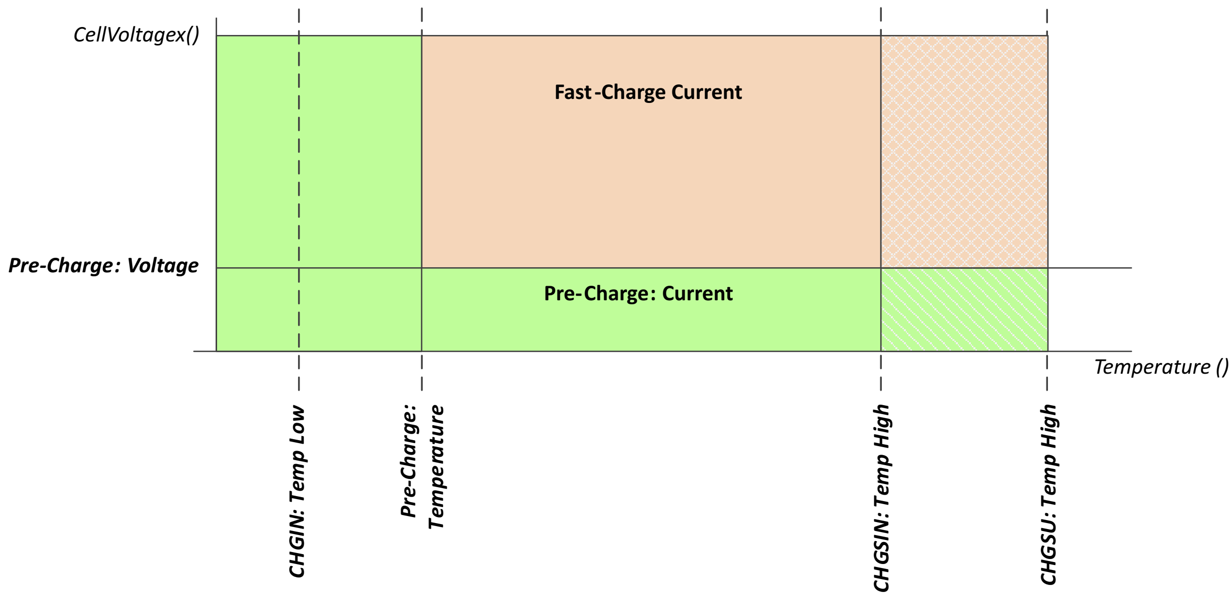
Class	Subclass	Name	Type	Min	Max	Default
Black Box	PF Status	1st PF Status 0–7	H2	0x0000	0xFFFF	0
Black Box	PF Status	1st PF Status 8–15	H2	0x0000	0xFFFF	0
Black Box	PF Status	1st PF Status 16–23	H2	0x0000	0xFFFF	0
Black Box	PF Status	1st PF Status 24–31	H2	0x0000	0xFFFF	0
Black Box	PF Status	1st Time to Next Event	U1	0	255	0
Black Box	PF Status	2nd PF Status 0–8	H2	0x0000	0xFFFF	0
Black Box	PF Status	2nd PF Status 9–15	H2	0x0000	0xFFFF	0
Black Box	PF Status	2nd PF Status 16–23	H2	0x0000	0xFFFF	0
Black Box	PF Status	2nd PF Status 24–32	H2	0x0000	0xFFFF	0
Black Box	PF Status	2nd Time to Next Event	U1	0	255	0

Class	Subclass	Name	Type	Min	Max	Default
Black Box	PF Status	3rd PF Status 0–8	H2	0x0000	0xFFFF	0
Black Box	PF Status	3rd PF Status 9–15	H2	0x0000	0xFFFF	0
Black Box	PF Status	3rd PF Status 16–23	H2	0x0000	0xFFFF	0
Black Box	PF Status	3rd PF Status 24–32	H2	0x0000	0xFFFF	0
Black Box	PF Status	3rd Time to Next Event	U1	0	255	0

Charge Algorithm

5.1 Introduction

The device can change the values of *ChargingVoltage()* and *ChargingCurrent()* based on *Temperature()*, *Cell Voltage1..15()* and system fault conditions. The *ChargingStatus()* register shows the state of the charging algorithm.



5.2 Fast and Pre-Charging

The charging algorithm adjusts *ChargingCurrent()* and *ChargingVoltage()* to allow the appropriate charging conditions to be read.

Current State	Condition	Action
Fast Charging	*Temperature()* > ***Precharge Temp*** + ***Hysteresis Temp*** AND ALL *CellVoltages1..15()* > Pre-Charging: Recovery Voltage AND *GaugingStatus() [EDV0]* = 0	*ChargingStatus()[FCHG]* = 1 *ChargingStatus()[PCHG]* = 0 *ChargingVoltage()* = ***Fast Charging: Voltage*** *ChargingCurrent()* = ***Fast Charging: Current***
Pre-Charging	*Temperature()* ≤ Pre-Charging: ***Precharge Temp*** + ***Hysteresis Temp*** OR ANY *CellVoltages1..15()* ≤ Pre-Charging: Start Voltage OR *GaugingStatus() [EDV0]* = 1	*ChargingStatus()[FCHG]* = 0 *ChargingStatus()[PCHG]* = 1 *ChargingVoltage()* = ***Fast Charging: Voltage*** *ChargingCurrent()* = ***Pre-Charging: Current***

Depending on the ***FET Options[PCHG_EN]*** settings, the external precharge FET or CHG FET can be used in PRE-CHARGE mode. Setting the ***Pre-Charging Current*** = 0 mA disables the precharge function by requesting 0 mA charging current from the charger.

FET Options[PCHG_EN]	FET Used
0	CHG
1	PCHG

Class	Subclass	Name	Type	Min	Max	Default	Unit
Charge Algorithm	Fast Charging	Voltage	I2	0	5000	4200	mV
Charge Algorithm	Fast Charging	Current	I2	0	32767	3000	mA
Charge Algorithm	Pre-Charging	Current	I2	0	32767	100	mA
Charge Algorithm	Pre-Charging	Start Voltage	I2	0	32767	2500	mV
Charge Algorithm	Pre-Charging	Recovery Voltage	I2	0	32767	2900	mV

5.3 Valid Charge Termination

The charge termination condition must be met to enable valid charge termination. The device has the following actions at charge termination, based on the flags settings:

- If ***FETOption[CHGFET] = 1***, CHG FET turns off.
- If ***CEDV Gauging Configuration[CSYNC]*** = 1, *RemainingCapacity()* = *FullChargeCapacity()*.
- If ***SBS Gauging Configuration[RSOCL]*** = 1, *RelativeStateOfCharge()* and *RemainingCapacity()* are held at 99% until charge termination occurs. Only on entering charge termination is 100% displayed.
- If ***SBS Gauging Configuration[RSOCL]*** = 0, *RelativeStateOfCharge()* and *RemainingCapacity()* are not held at 99% until charge termination occurs. Fractions of % greater than 99% are rounded up to display 100%.

Status	Condition	Action
Charging	*GaugingStatus()[REST]* = 0 AND *GaugingStatus()[DSG]* = 0	Charge Algorithm active
Valid Charge Termination	All of the following conditions must occur for two consecutive 40-s periods: Charging (that is, *BatteryStatus[DSG]* = 0) AND *AverageCurrent()* < ***Charge Term Taper Current*** AND Max (*CellVoltage1..15()*) + ***Charge Term Voltage*** ≥ *ChargingVoltage()*/number of cells in series AND The accumulated change in capacity > 0.25 mAh since current and voltage termination conditions where first detected.	*ChargingStatus()[VCT]* = 1 *ChargingVoltage()* = Charging Algorithm *ChargingCurrent()* = Charging Algorithm *BatteryStatus()[FC]* = 1 and *GaugingStatus()[FC]* = 1 if ***SOCFlagConfig A[FCSETVCT]*** = 1 *BatteryStatus()[TCA]* = 1 and *GaugingStatus()[TCA]* = 1 if ***SOCFlagConfig B[TCSETVCT]*** = 1

Class	Subclass	Name	Type	Min	Max	Default	Unit
Charging Algorithms	Termination Config	Charge Term Taper Current	I2	0	32767	250	mA
Charging Algorithms	Termination Config	Charge Term Voltage	I2	0	32767	75	mV

5.4 Charge and Discharge Alarms

The *[TCA]* and *[FC]* bits in *BatteryStatus()* can be set at charge termination as well as based on RSOC when the device is in CHARGE state (that is, *BatteryStatus[DSG]* = 0). If more than one set and clear conditions are selected, then the corresponding flag will be set whenever a valid set or clear condition is met. The same functionality is applied to the *[TDA]* and *[FD]* bits in *BatteryStatus()*.

Per the *Smart Battery Data Specification v1.1*, TDA is only active while discharging and TCA is only active while charging but the BQ78350-R1A will only follow this particular requirement if ***SOC Flag Config [SBS_COMP]*** = 1. By default, the TCA and TDA flags will not change based on current magnitude or direction.

NOTE: In *BatteryStatus()*, the *[TCA]* bit, as well as the *[TDA]* and *[FD]* bits, are also set and cleared based on safety and permanent fail protections. In *GaugingStatus()*, however, these bits do not react on the safety protections.

GaugingStatus[TC][TD][FC][FD] are the status flags based on the gauging conditions only. These flags are set and cleared based on ***SOC Flag Config***.

The *GaugingStatus[TC][TD]* flags are not the same as the *BatteryStatus[TCA][TDA]* flags. The *[TCA]* and *[TDA]* flags can be set or cleared by the gauging event or by the safety or PF events. These flags also clear if charging current is not present. The *[TC]* and *[TD]* flags, however, only set and clear by a gauging event.

GaugingStatus[FC][FD] has the same behavior as *BatteryStatus[FC][FD]*.

The table below summarizes the various options to set and clear the *[TC]* and *[FC]* flags in *GaugingStatus()*.

Flag	Set Criteria	Set Condition	Enable
[TC]	RSOC	*RelativeStateOfCharge()* ≥ ***TC: Set % RSOC Threshold***	***SOC Flag Config [TCSetRSOC]*** = 1
	Valid Charge Termination (enable by default)	When *ChargingStatus[VCT]* = 1	***SOC Flag Config [TCSetVCT]*** = 1
[FC]	RSOC	*RelativeStateOfCharge()* ≥ ***FC: Set % RSOC Threshold***	***SOC Flag Config [FCSetRSOC]*** = 1
	Valid Charge Termination (enable by default)	When *ChargingStatus[VCT]* = 1	***SOC Flag Config [FCSetVCT]*** = 1

Flag	Clear Criteria	Clear Condition	Enable
[TC]	RSOC (enable by default)	*RelativeStateOfCharge()* ≤ ***TC: Clear % RSOC Threshold***	***SOC Flag Config [TCClearRSOC]*** = 1
[FC]	RSOC (enable by default)	*RelativeStateOfCharge()* ≤ ***FC: Clear % RSOC Threshold***	***SOC Flag Config [FCClearRSOC]*** = 1

The tables below summarizes the various options to set and clear the ***[TD]*** and ***[FD]*** flags in both *BatteryStatus()* and *GaugingStatus()*.

Flag	Set Criteria	Set Condition	Enable
[TD]	RSOC (enable by default)	*RelativeStateOfCharge()* ≤ ***TD: Set % RSOC Threshold***	***SOC Flag Config [TDSetRSOC]*** = 1
[FD]	RSOC (enable by default)	*RelativeStateOfCharge()* ≤ ***FD: Set % RSOC Threshold***	***SOC Flag Config [FDSetRSOC]*** = 1

Flag	Clear Criteria	Clear Condition	Enable
[TD]	RSOC (enable by default)	*RelativeStateOfCharge()* ≥ ***TD: Clear % RSOC Threshold***	***SOC Flag Config [TDClearRSOC]*** = 1
[FD]	RSOC (enable by default)	*RelativeStateOfCharge()* ≥ ***FD: Clear % RSOC Threshold***	***SOC Flag Config [FDClearRSOC]*** = 1

The SOC configuration is stored in the following data flash.

Table 5-1. SOC FLAG Config

Class	Subclass	Name	Format	Size in Bytes	Min	Max	Default	Unit
Settings	Configuration	SOC Flag Config	Hex	2	0x0000	0xFFFF	0x02FB	

15	14	13	12	11	10	9	8
RSVD	RSVD	RSVD	SBS_COMP	RSVD	RSVD	FCCLEARRSOC	FCSETRSOC
7	6	5	4	3	2	1	0
FDCLEARRSOC	FDSETRSOC	TCSETVCT	FCSETVCT	TCCLEARRSOC	TCSETRSOC	TDCLEARRSOC	TDSETRSOC

RSVD (Bits 15–13): Reserved

SMB_COMP (Bit 12): Enable SOC FLAG Smart Battery Standard specification compliance

1 = Enabled
0 = Disabled (default)

RSVD (Bits 11–10): Reserved

FCCLEARRSOC (Bit 9): Enable FC flag clear by RSOC threshold

1 = Enabled (default)
0 = Disabled

FCSETRSOC (Bit 8): Enable FC flag set by RSOC threshold

1 = Enabled
0 = Disabled (default)

FDCLEARRSOC (Bit 7): Enable TC flag set by primary charge

1 = Enabled (default)
0 = Disabled

FDSETRSOC (Bit 6): Enable FD flag set by RSOC threshold

1 = Enabled (default)
0 = Disabled

TCSETVCT (Bit 5): Enable TC flag set by primary charge

1 = Enabled (default)
0 = Disabled

FCSETVCT (Bit 4): Enable FC flag set by primary charge

1 = Enabled (default)
0 = Disabled

TCCLEARRSOC (Bit 3): Enable TC flag clear by RSOC threshold

1 = Enabled (default)
0 = Disabled

TCSETRSOC (Bit 2): Enable TC flag set by RSOC threshold

1 = Enabled
0 = Disabled (default)

TDCLEARRSOC (Bit 1): TDCLEARRSOC—Enable TD flag clear by RSOC threshold

1 = Enabled (default)
0 = Disabled

TDSETRSOC (Bit 0): TDSETRSOC—Enable TD flag set by RSOC threshold

1 = Enabled (default)
0 = Disabled

Class	Subclass	Name	Type	Min	Max	Default
Fuel Gauging	FD	Set RSOC % Threshold	U1	0%	100%	0%
Fuel Gauging	FD	Clear RSOC % Threshold	U1	0%	100%	5%
Fuel Gauging	FC	Set RSOC % Threshold	U1	0%	100%	100%
Fuel Gauging	FC	Clear RSOC % Threshold	U1	0%	100%	95%
Fuel Gauging	TD	Set RSOC % Threshold	U1	0%	100%	6%
Fuel Gauging	TD	Clear RSOC % Threshold	U1	0%	100%	8%
Fuel Gauging	TC	Set RSOC % Threshold	U1	0%	100%	100%
Fuel Gauging	TC	Clear RSOC % Threshold	U1	0%	100%	95%

5.5 Charge Disable

The device can disable charging if certain safety conditions are detected setting the *OperationStatus()[XCHG]* = 1.

Status	Condition	Action
Normal	*ALL PFStatus()* = 0 AND *SafetyStatus()[COV]* = 0 AND *SafetyStatus()[OTC]* = 0 AND *SafetyStatus()[UTC]* = 0 AND *SafetyStatus()[OCC]* = 0 AND *SafetyStatus()[CTO]* = 0 AND *SafetyStatus()[PTO]* = 0 AND *GaugingStatus()[TCA]* = 0 if ***FET Options[CHGFET]*** = 1	*ChargingVoltage()* = Charging Algorithm *ChargingCurrent()* = Charging Algorithm
Trip	*ANY PFStatus()* = 1 OR *SafetyStatus()[COV]* = 1 OR *SafetyStatus()[OTC]* = 1 OR *SafetyStatus()[UTC]* = 1 OR *SafetyStatus()[OCC]* = 1 OR *SafetyStatus()[CTO]* = 1 OR *SafetyStatus()[PTO]* = 1 OR *GaugingStatus()[TCA]* = 1 if ***FET Options[CHGFET]*** = 1	*ChargingVoltage()* = 0 *ChargingCurrent()* = 0

5.6 Charge Inhibit

The device can inhibit the start of charging at high and low temperatures to prevent damage of the cells. This feature prevents the start of charging when the temperature is at the inhibit range; therefore, if the device is already in the charging state when the temperature reaches the inhibit range, a FET action will not take place even if ***FET Options[CHGIN]*** = 1.

Status	Condition	Action
Normal	*BatteryStatus()[DSG]* = 0 ***Charge Inhibit/Charge Suspend Low Temp*** + ***Hysteresis Temp*** < *Temperature()* < ***Charge Inhibit High Temp*** – ***Hysteresis Temp***	*ChargingStatus()[IN]* = 0 *ChargingVoltage()* = charging algorithm *ChargingCurrent()* = charging algorithm
Trip	*BatteryStatus()[DSG]* = 0 ***Charge Inhibit/Suspend Low Temp*** > *Temperature()* > ***Charge Inhibit Temp High***	*ChargingStatus()[IN]* = 1 *ChargingVoltage()* = 0 *ChargingCurrent()* = 0 No charging is allowed if ***FET Options[CHGIN]*** = 1.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Charge Algorithm	Temperature Ranges	Charge Inhibit/Suspend Low Temp	I1	–127	128	0	°C
Charge Algorithm	Temperature Ranges	Pre-Charge Temp	I1	–127	128	12	°C
Charge Algorithm	Temperature Ranges	Charge Inhibit High Temp	I1	–127	128	45	°C
Charge Algorithm	Temperature Ranges	Hysteresis Temp	I1	–127	128	3	°C

5.7 Charge Suspend

The device can suspend charging at high and low temperatures to prevent damage of the cells. Care should be taken to ensure Charge Inhibit and Charge Suspend features are configured correctly as upon Charge Suspend detection [CHGSU= 1], then Charge Inhibit detection criteria will have to be passed prior to restarting charge.

Status	Condition	Action
Normal	*BatteryStatus()[DSG]* = 0 ***Charge Inhibit/Suspend Low Temp*** < *Temperature()* < ***Charge Suspend High Temp***	*ChargingStatus()[SU]* = 0 *ChargingVoltage()* = charging algorithm *ChargingCurrent()* = charging algorithm
Trip	*BatteryStatus()[DSG]* = 0 ***Charge Inhibit/Suspend Low Temp*** > *Temperature()* > ***Charge Suspend High Temp***	*ChargingStatus()[SU]* = 1 *ChargingVoltage()* = 0 *ChargingCurrent()* = 0 No charging is allowed if ***FET Options[CHGSU]*** = 1.

Class	Subclass	Name	Type	Min	Max	Default	Unit
Charge Algorithm	Temperature Ranges	Charge Suspend High Temp	I1	–127	128	55	°C

System Present

6.1 Introduction

The BQ78350-R1A has the capability to detect the presence of a system and/or a charger through the state of the $\overline{\text{PRES}}$ pin. This can be used to disable the battery output when the BQ78350-R1A detects the battery has been removed from the system or charger.

6.2 System Present Detection and Action

The $\overline{\text{PRES}}$ pin is polled every 250 ms and if it is detected High for four consecutive 250-ms samples, then the CHG, DSG, and PCHG FETs are turned off. If $\overline{\text{PRES}}$ is detected Low, then the FETs are allowed to be turned on depending on other safety and charging related algorithms. If this feature is not required, then the $\overline{\text{PRES}}$ pin should be tied to VSS.

System Present Detection can be selected as an action to clear some types of protection faults. See *Enabled Removal Recovery*.